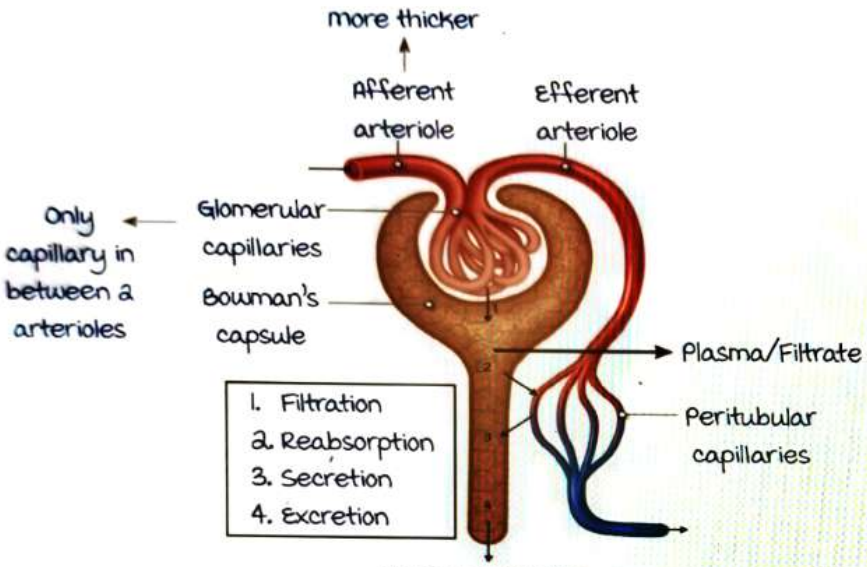
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GLOMERULAR FILTRATION RATE

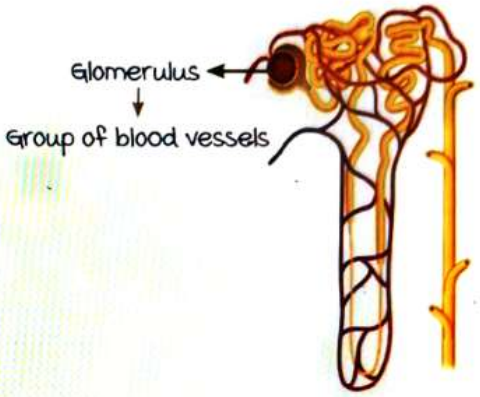
Glomerular Capillaries and GFR

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Excretion = Filtration − Reabsorption + Secretion

Renal physiology



Nephron

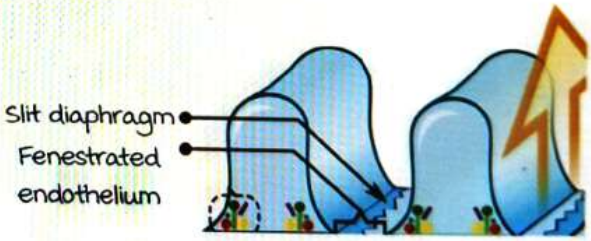
- Functional unit of kidney.
- 1 million/kidney.
- Loss of nephron → Other nephrons compensate.

ULTRAFILTRATE:

- Filtered fluid: Ultrafiltrate (Proteins are not filtered, only plasma).
- Normal GFR: 125 mL/min or 180 L/day.

Reason for proteins not being filtered:

1. Heparan sulphate:
   - Line glomerular capillaries.
   - Negatively charged & repel proteins (Anions, -ve).
   - In renal disease: Proteins filtered (D/t loss of heparan sulphate).



2. Fenestrated endothelium:
   - Small gap between cells (4–8 mm).
   - Restrict protein passage.
3. Slit diaphragm:
   - Foot process → Reinforce the fenestrations → Smaller gaps → Slit diaphragm of podocytes.
   - Restrict protein passage.

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- Integrity of slit diaphragm is maintained by :

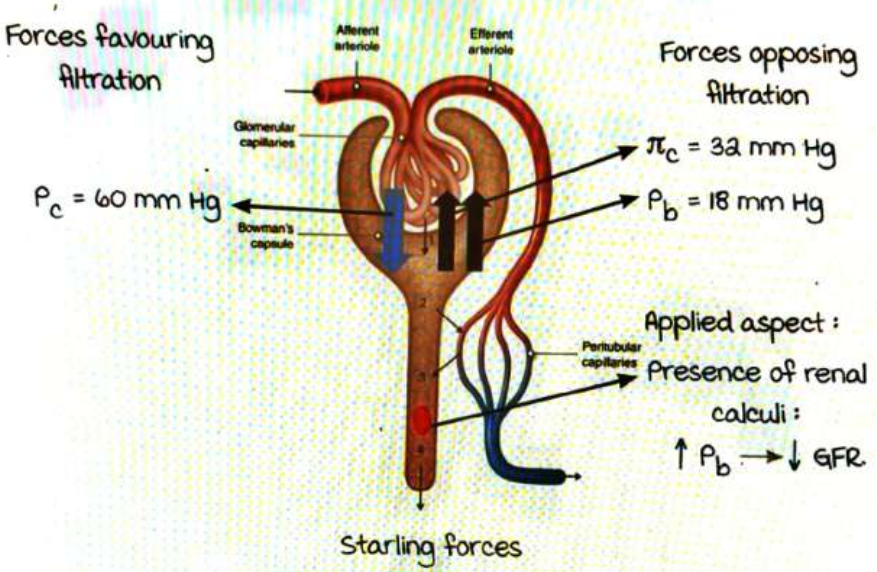
Protein	Mutation (Leads to protein excretion)
1. Nephrin	Congenital finnish type nephrotic syndrome (NPHS I gene)
2. Podocin	Steroid resistant nephrotic syndorme
3. α-actinin	Focal segmental glomerulosclerosis (FSGS).

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FILTRATION FORCES :

Starling forces :

- Hydrostatic pressure in glomerular capillaries (P_c).
- Oncotic pressure in glomerular capillary (π_c).
- Hydrostatic pressure in Bowman's capsule (P_b).



Net filtration pressure :

= Forces that favour filtration – forces that oppose filtration.

= $P_c - (\pi_c + P_b)$.

= 60 – (32 + 18).

= 10 mm Hg.

MEASUREMENT OF GFR :

Criteria of a substance used to measure GFR :

- Freely filtered.
- No reabsorption.
- No secretion.
- Non toxic.

Substances :

1. Inulin

Feedback

Inulin clearance rate by normal kidneys = 125 mL/min = Normal GFR.

Gold standard (Inulin filtered = Inulin excreted).

Procedure : Inulin clearance.

IV inulin → Plasma —Filtered→ Kidneys —Excreted→ Urine.

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Filtration x plasma concentration (P) = urine concentration (U) x volume (V).

GFR x P = U x V.

GFR = UV/P ; ↑ P → ↓ GFR.

2. Creatinine :

- Plasma creatinine is used for estimation in clinical practice.
- ↑ plasma creatinine → ↓ GFR.

3. Cystatin C :

- Newer marker for GFR.

Applied aspect :

Relation with GFR	Substance
Clearance = GFR	Inulin
Clearance > GFR (New net secretion)	Para amino hippuric acid (PAH)
Clearance < GFR (New net reabsorption)	Glucose

Renal Circulation

00:29:04

Renal blood flow (RBF) :

1250 mL/min (20-25% of cardiac output).

Renal plasma flow :

- 55% of RBF = 625 mL/min.
- measurement :
  - Substance used : PAH.
  - Formula :

$$\text{Effective/estimated RPF} = \frac{U_{PAH} \times V}{P_{PAH}}$$

$$\text{True RPF} = \frac{\text{Effective RPF}}{\text{Extraction ratio}} = \frac{625}{0.9} = 700 \text{ mL/min}$$

Extraction ratio is % of PAH excreted by kidneys = 90%.

Filtration fraction :

- $\frac{\text{GFR}}{\text{Renal plasma flow}} = \frac{125}{625} = 20\%$
- 1/5 th of renal plasma is filtered at any point of time.

Feedback

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Regulation of GFR & RBF :

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1. Constriction of afferent & efferent arteriole :

Substance	Effect on arteriole/capillaries	Effect on GFR	Effect on RBF
Norepinephrine (In shock, sympathetic stimulation)	Constriction of afferent arteriole → ↓ blood flow to glomerular capillaries → ↓ P_c	↓	↓
Angiotensin II	Constriction of efferent arteriole : Biphasic response 1. Initial phase (Stronger) Blood pooling in glomerular capillaries → ↑ P_c → ↑GFR. 2. Later phase ↑ protein in glomerular capillaries → ↑ π_c → ↓ GFR.	↑ GFR ↓ Later ↓ GFR	↓ (In both phases).
PGE2 (AKA renal prostaglandin) : Produced by renal medullary interstitial cells (RMIC)	Dilatation of afferent arteriole : ↑ blood flow to renal cortex ↓ blood flow to renal medulla	↑	↑
ACE inhibitors	Dilatation of efferent arteriole : Blood moves quickly from glomerular capillaries → ↓P_c	↓	↑

Note :

Efferent arteriolar constriction ⟶ ↑ π_c in peritubular capillaries.

2. Autoregulation :

Despite perfusion pressure change ⟶ No change in GFR & RBF.

Occurs between 80–180 mm Hg.

Doesn't need neural influence (Occurs in denervated kidney).

Mechanism :

- myogenic : Vascular.
  - ↑ RBF → Vessel wall stretch → Opens Ca^{2+} channels → Ca^{2+} influx → Vasoconstriction → ↓ RBF.
- Tubulo glomerulo feedback (TGF) :
  - Negative feedback mechanism.
  - ↑ GFR → ↑NaCl filtration → Activates macula densa (GFR sensor) → Release adenosine → Constricts afferent arteriole → ↓ GFR.
  - Applied aspect :
    a. Proteins require Na^+ for their absorption.
    b. High protein diet → Na^+ reaches macula densa → ↑ GFR.

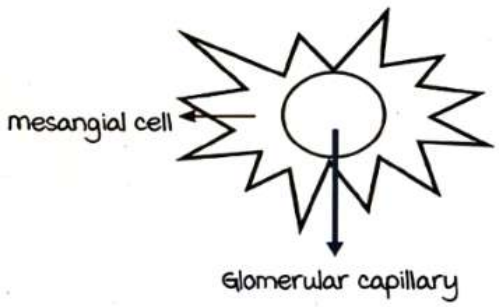
Feedback

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3. Mesangial cell contraction & relaxation :

↓ GFR	↑ GFR
Endothelin, angiotensin II, norepinephrine	NO, dopamine, cAMP, natriuretic peptide
↓	↓
mesangial cell contraction	mesangial cell relaxation
↓	↓
↓ blood flow to capillaries	Expansion of capillaries
↓	↓
↓ GFR	↑ RBF & ↑ GFR



Feedback